Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *BLURB* | ***Context:*** *EU/French Pharmaceutical Regulatory NLP — BLURB Assessment*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's annotator population is biomedical-research-oriented and heterogeneous: Amazon Mechanical Turkers and Upwork contributors with medical training for EBM PICO [Q60], 'five expert-level annotators' for BIOSSES [Q74], 'biomedical experts' for BioASQ [Q88], and unspecified/varied for the remaining datasets. None are regulatory affairs specialists or legal experts operating under EMA/ANSM normative frameworks. The deployment explicitly anticipates that biomedical NLP annotators will 'prioritize clinical relevance' while regulatory ground truth requires 'rigid legal constraint satisfaction,' producing systematic disagreements on borderline cases (elicitation A4). Web-confirmed: no published IAA study comparing biomedical vs regulatory annotators exists, and no regulatory-expert-annotated French pharmaceutical NER/STS dataset has been identified [WEB-12]. BLURB also does not systematically report IAA, annotator demographics, or annotation protocols across datasets. ChemProt label expansion via assigning false labels to unannotated pairs [Q66] further reflects research-pragmatic conventions that may be inappropriate where unannotated content has legal-evidentiary weight. Some annotation quality signals (BIOSSES expert annotators, EBM PICO test set quality stratification) [Q60, Q74] exist, hence score 2 rather than 1.

ID	Evidence
Q60	"The training and development sets were labeled by Amazon Mechanical Turkers, whereas the test set was labeled by Upwork contributors with prior medical training." (p.8)
Q66	"The ChemProt annotation is not exhaustive for all chemical-protein pairs. Following previous work [34, 45], we expand the training and development sets by assigning a false label for all chemical-protein pairs that occur in a training or development sentence, but do not have an explicit label in the ChemProt corpus." (p.8)
Q74	"The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)." (p.9)
Q88	"The BioASQ corpus [42] contains multiple question answering tasks annotated by biomedical experts, including yes/no, factoid, list, and summary questions." (p.9)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)

Strengths

- BIOSSES uses five expert annotators with documented inter-annotator process [Q74], offering a usable methodological template for setting up regulatory-expert annotation
- BLURB documents the EBM PICO quality stratification (MTurk train/dev vs medically trained Upwork test) [Q60], demonstrating awareness that annotator expertise affects label quality
- BioASQ annotations by 'biomedical experts' [Q88] provide a partial precedent for domain-expert annotation in QA construction

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Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels do not reflect regional regulatory stakeholder perspectives. Annotators are biomedical/clinical professionals [Q60, Q74, Q88], not regulatory affairs specialists or EMA/ANSM-aligned legal experts.

OC-2: Disagreement is explicitly anticipated by the deployment elicitation (A4) on borderline cases, where biomedical annotators prioritize clinical relevance while regulatory ground truth prioritizes legal constraint satisfaction. No empirical IAA study addresses this gap [WEB-12].

OC-3: Annotator demographics are unevenly documented: EBM PICO sources are named [Q60], BIOSSES annotator count and expertise level given [Q74], BioASQ described generically as 'biomedical experts' [Q88]; for most datasets (BC5, NCBI-Disease, BC2GM, JNLPBA, ChemProt, DDI, GAD, HoC, PubMedQA) the paper does not provide annotator demographics or protocols. INSUFFICIENT DOCUMENTATION across most of the benchmark.

OC-4: Re-annotation by EMA/ANSM-trained regulatory affairs specialists would be required for any deployment use, but is not feasible without a French regulatory document corpus to begin with.

OC-5: BLURB uses macro-averaging across task types [Q52, Q150] and averages across multiple runs for small datasets [Q139, Q140] but does not document any aggregation method that preserves minority annotator perspectives — particularly relevant for the BIOSSES five-annotator average [Q75], which collapses disagreement into a single regression target.

OC-6: Convergent validity is harmed (labels do not correlate with regulatory-stakeholder judgments); external validity is harmed (BLURB performance is unlikely to predict performance against EMA/ANSM ground truth).

ID	Evidence
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
Q60	“The training and development sets were labeled by Amazon Mechanical Turkers, whereas the test set was labeled by Upwork contributors with prior medical training.” (p.8)
Q74	“The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings).” (p.9)
Q75	“It is a regression task, with the average score as the final annotation.” (p.9)
Q88	“The BioASQ corpus [42] contains multiple question answering tasks annotated by biomedical experts, including yes/no, factoid, list, and summary questions.” (p.9)
Q139	“Due to random initialization of the task-specific model and drop out, the performance may vary for different random seeds, especially for small datasets like BIOSSES, BioASQ, and PubMedQA.” (p.13)
Q140	“We report the average scores from ten runs for BIOSSES, BioASQ, and PubMedQA, and five runs for the others.” (p.13)

Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)
WEB-10	EMA 2024 AI Observatory Report — internal regulatory NLP tools developed under EMA staff oversight, not represented in any public benchmark (https://www.ema.europa.eu/en/documents/report/2024-ai-observatory-report_en.pdf)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)

Information Gaps

- Empirical magnitude of disagreement between biomedical and regulatory annotators on shared regulatory text — no IAA study exists [WEB-12]

Requires Expert Verification

- Whether existing EMA/ANSM internal review checklists could be operationalized as annotation guidelines for a regulatory-grade gold standard

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Remediation

Gap 1: No BLURB annotators are regulatory affairs specialists or EMA/ANSM-aligned legal experts; systematic disagreement on borderline cases is anticipated (elicitation A4).

Recommendation: Establish a regulatory-expert annotation pool drawn from regulatory affairs and pharmacovigilance specialists, with annotation guidelines anchored in EMA QRD templates [WEB-1] and ANSM procedural documentation. Run a calibrated IAA study comparing regulatory-expert vs biomedical-NLP annotator labels on a shared sample of regulatory text to quantify the gap.

ID	Evidence
WEB-1	EMA QRD SmPC Template v10.4 (Feb 2024) — defines the formulaic template language the deployment must process (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)